

第六次习题课解答 二重积分及计算

1. 求解下列各题:

(1) 求极限: $\lim_{n \rightarrow \infty} \sum_{j=1}^n \sum_{i=1}^n \frac{n}{(n+i)(n^2+j^2)}.$

解:
$$\lim_{n \rightarrow \infty} \sum_{j=1}^n \sum_{i=1}^n \frac{n}{(n+i)(n^2+j^2)} = \lim_{n \rightarrow \infty} \sum_{j=1}^n \sum_{i=1}^n \frac{1}{(1+\frac{i}{n})(1+(\frac{j}{n})^2)} \frac{1}{n^2}$$
$$= \iint_{\substack{0 \leq x \leq 1 \\ 0 \leq y \leq 1}} \frac{1}{(1+x)(1+y^2)} dx dy = \int_0^1 \frac{1}{1+x} dx \int_0^1 \frac{1}{1+y^2} dy = \frac{\pi}{4} \ln 2.$$

(2) 求 $f(x) = \int_1^x \sin t^2 dt$ 在 $[0,1]$ 上的平均值, 即求 $\int_0^1 f(x) dx$.

解:
$$\int_0^1 f(x) dx = \int_0^1 \left(\int_1^x \sin t^2 dt \right) dx = - \int_0^1 \left(\int_x^1 \sin t^2 dt \right) dx$$
$$= - \int_0^1 \left(\int_0^t \sin t^2 dx \right) dt = - \int_0^1 t \sin t^2 dt = \frac{1}{2} (\cos 1 - 1).$$

(3) 当 $t \rightarrow 0^+$ 时, 求无穷小量 $f(t) = \iint_{x^2+y^2 \leq t^2} [1 - \cos(x^2 + y^2)] dx dy$ 的阶.

解: 因为 $f(t) = \iint_{x^2+y^2 \leq t^2} [1 - \cos(x^2 + y^2)] dx dy = \int_0^{2\pi} \int_0^t (1 - \cos r^2) r dr d\theta$
$$= \pi(t^2 - \sin t^2) = \pi(t^2 - t^2 + \frac{1}{6}t^6 + o(t^6)) = \pi(\frac{1}{6}t^6 + o(t^6)), t \rightarrow 0,$$

因此当 $t \rightarrow 0^+$ 时, $f(t)$ 是 6 阶无穷小量.

(4) 令 $D = \{(x, y) | x^2 + y^2 \leq 1, x \geq 0\}$. 计算 $\iint_D \frac{1+xy}{1+x^2+y^2} dx dy$.

解: 因为积分区域 D 关于 x 轴对称, 且 $\frac{xy}{1+x^2+y^2}$ 是 y 的奇函数, 因此

$$\iint_D \frac{xy}{1+x^2+y^2} dx dy = 0.$$

故
$$\iint_D \frac{1+xy}{1+x^2+y^2} dx dy = \iint_D \frac{1}{1+x^2+y^2} dx dy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^1 \frac{r}{1+r^2} dr d\theta = \frac{\pi}{2} \ln 2.$$

(5) 设 $F(t) = \int_0^t dx \int_x^t e^{x+y} \cos \sqrt{y} dy$ ($t > 0$), 求 $F'(t)$.

解: $F(t) = \int_0^t dx \int_x^t e^{x+y} \cos \sqrt{y} dy = \int_0^t e^y \cos \sqrt{y} dy \int_0^y e^x dx = \int_0^t e^y \cos \sqrt{y} (e^y - 1) dy$,

故 $F'(t) = e^t \cos \sqrt{t} (e^t - 1)$.

(6) 设 $f(x, y)$ 为连续函数且 $f(x, y) = f(y, x)$. 证明:

$$\int_0^1 dx \int_0^x f(x, y) dy = \int_0^1 dx \int_0^{1-x} f(1-x, 1-y) dy.$$

证: 令 $x = 1-u, y = 1-v$, 则 $0 \leq v \leq 1, 0 \leq u \leq v$, 且 $\left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| = 1$. 于是

$$\int_0^1 dx \int_0^x f(1-x, 1-y) dy = \int_0^1 dv \int_0^v f(u, v) du = \int_0^1 dv \int_0^v f(v, u) du = \int_0^1 dx \int_0^x f(x, y) dy.$$

(7) 将定积分 $\int_0^1 \frac{\ln(1+x)}{(2-x)^2} dx$ 转化为二重积分计算.

$$\begin{aligned} \text{解: } \int_0^1 \frac{\ln(1+x)}{(2-x)^2} dx &= \int_0^1 \left(\frac{1}{(2-x)^2} \int_0^x \frac{1}{1+y} dy \right) dx = \int_0^1 \frac{1}{1+y} \left(\int_y^1 \frac{1}{(2-x)^2} dx \right) dy \\ &= \int_0^1 \frac{1}{1+y} \left(1 - \frac{1}{2-y} \right) dy = \frac{1}{3} \int_0^1 \left(\frac{2}{1+y} - \frac{1}{2-y} \right) dy = \frac{1}{3} \ln 2. \end{aligned}$$

(8) 设 $f(x) \in C[0, 1]$. 证明: $\int_0^1 e^{f(x)} dx \int_0^1 e^{-f(x)} dx \geq 1$.

证明: 令 $D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$, 则积分区域 D 关于直线 $y = x$ 对称, 因此由轮换对称性,

$$\begin{aligned} \int_0^1 e^{f(x)} dx \int_0^1 e^{-f(x)} dx &= \int_0^1 e^{f(x)} dx \int_0^1 e^{-f(y)} dy = \iint_D e^{f(x)-f(y)} dx dy = \iint_D e^{f(y)-f(x)} dx dy \\ &= \frac{1}{2} \iint_D (e^{f(x)-f(y)} + e^{f(y)-f(x)}) dx dy \geq 1. \end{aligned}$$

2. 解答下列各题:

(1) 设 $\Omega \subset \mathbb{R}^3$ 是由锥面 $z = 1 - \sqrt{x^2 + y^2}$ 以及平面 $z = x$ 和 $x = 0$ 围成, 求空间区域 Ω 的体积.

解: 空间区域 Ω 在 xoy 坐标平面内的投影区域 D 由平面曲线 $1-x = \sqrt{x^2 + y^2}$ 以及直线

$x = 0$ 围成, 且 D 在极坐标系下表示为 $D = \left\{ (r, \theta) \mid -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, 0 \leq r \leq \frac{1}{1+\cos \theta} \right\}$, 因

此空间区域 Ω 的体积

$$\begin{aligned}
 V(\Omega) &= \iint_D (1 - \sqrt{x^2 + y^2} - x) dx dy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{\frac{1}{1+\cos\theta}} (1 - r(1 + \cos\theta)) r dr \\
 &= \frac{1}{6} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{(1 + \cos\theta)^2} d\theta = \frac{1}{12} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{\cos^4 \theta} d\theta = \frac{1}{12} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\frac{1}{\cos^2 \theta} + \frac{\tan^2 \theta}{\cos^2 \theta} \right) d\theta = \frac{2}{9}.
 \end{aligned}$$

(2) 求曲线 $(x^2 + y^2)^2 = 2ax^3$ 所围平面区域的面积.

解: 由 $(x^2 + y^2)^2 = 2ax^3$, 得 $r = 2a \cos^3 \theta$ ($-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$), 所求面积

$$S = \iint_D dx dy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2a \cos^3 \theta} \rho d\rho = 2a^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^6 \theta d\theta = \frac{5}{8} \pi a^2.$$

(3) 分别求出由平面 $z = x - y$, $z = 0$ 与圆柱面 $x^2 + y^2 = 2x$ 所围成的两个空间几何体的体积.

解: 在极坐标系下, 记 $D_1 = \{(r, \theta) | 0 \leq r \leq 2 \cos \theta, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{4}\}$, 且

$$D_2 = \{(r, \theta) | 0 \leq r \leq 2 \cos \theta, \frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}\}.$$

则所围空间几何体位于 Oxy 平面上方部分的体积

$$\begin{aligned}
 V_1 &= \iint_{D_1} (x - y) dx dy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{4}} d\theta \int_0^{2 \cos \theta} (\cos \theta - \sin \theta) r^2 dr \\
 &= \frac{8}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{4}} (\cos^4 \theta - \sin \theta \cos^3 \theta) d\theta = \frac{3}{4} \pi + \frac{5}{6}.
 \end{aligned}$$

位于 Oxy 平面下方部分的体积

$$\begin{aligned}
 V_2 &= - \iint_{D_2} (x - y) dx dy = - \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^{2 \cos \theta} (\cos \theta - \sin \theta) r^2 dr \\
 &= - \frac{8}{3} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cos^4 \theta - \sin \theta \cos^3 \theta) d\theta = \frac{5}{6} - \frac{\pi}{4}.
 \end{aligned}$$

(4) 求两个球体 $x^2 + y^2 + z^2 \leq 1$ 与 $x^2 + y^2 + (z - 2)^2 \leq 4$ 所围立体的体积.

解: 两个球面的交线方程为 $\begin{cases} x^2 + y^2 + z^2 = 1 \\ x^2 + y^2 + (z - 2)^2 = 4, \end{cases}$ 解之得 $\begin{cases} x^2 + y^2 = \frac{15}{16} \\ z = \frac{1}{4}. \end{cases}$ 于是所求立体

体积为

$$\begin{aligned}
\iint_{x^2+y^2 \leq 15/16} (z_1(x, y) - z_2(x, y)) dx dy &= \iint_{x^2+y^2 \leq 15/16} \left[\sqrt{1-x^2-y^2} - \left(2 - \sqrt{4-x^2-y^2} \right) \right] dx dy \\
&= \int_{x^2+y^2 \leq 15/16} \left(\sqrt{1-x^2-y^2} + \sqrt{4-x^2-y^2} \right) dx dy - 2\pi \cdot \frac{15}{16} \\
&= \int_0^{2\pi} d\theta \int_0^{\sqrt{15/4}} \left(\sqrt{1-r^2} + \sqrt{4-r^2} \right) r dr - \frac{15\pi}{8} \\
&= \pi \int_0^{15/16} (\sqrt{1-s} + \sqrt{4-s}) ds - \frac{15\pi}{8} = \frac{13\pi}{24}.
\end{aligned}$$

(5) 求由曲线 $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 = x^2 + y^2$ 所围成的平面图形的面积.

解: 令 $x = ar \cos \theta, y = br \sin \theta$, 则 $\left| \det \frac{\partial(x, y)}{\partial(r, \theta)} \right| = abr$, 曲线所围平面区域 D 在极坐标系

下的区域变换为 $D' = \{(r, \theta) : 0 \leq \theta \leq 2\pi, 0 \leq r \leq \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}\}$. 于是所求区域的面积

$$\begin{aligned}
S(D) &= \iint_D dx dy = \iint_{D'} abr dr d\theta = ab \int_0^{2\pi} d\theta \int_0^{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} r dr \\
&= \frac{1}{2} ab\pi(a^2 + b^2).
\end{aligned}$$

3. 通过适当的坐标变换, 计算下列二重积分.

(1) $I = \iint_D (\sqrt{x^2 + y^2} + y) dx dy$, 其中 D 是介于圆周 $x^2 + y^2 = 4$ 与圆周 $(x+1)^2 + y^2 = 1$ 之间的部分.

解: 积分区域 D 关于 x 轴对称, 故 $\iint_D y dx dy = 0$.

令 $D_1 = \{(x, y) | x^2 + y^2 \leq 4\}$, $D_2 = \{(x, y) | (x+1)^2 + y^2 \leq 1\}$, 则

$$\begin{aligned}
I &= \iint_D \sqrt{x^2 + y^2} dx dy = \iint_{D_1} \sqrt{x^2 + y^2} dx dy - \iint_{D_2} \sqrt{x^2 + y^2} dx dy \\
&= \int_0^{2\pi} d\theta \int_0^2 r^2 dr - \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} d\theta \int_0^{-2\cos\theta} r^2 dr = \frac{16}{9} (3\pi - 2).
\end{aligned}$$

(2) $\iint_D (\sqrt{x} + \sqrt{y}) dx dy$, D 是由 $\sqrt{x} + \sqrt{y} = 1$, $x = 0$, $y = 0$ 所围成的平面区域.

解: 令 $\begin{cases} x = u^2 \\ y = v^2 \end{cases}$, 则积分区域 D 在新坐标系下变换为 $\Omega = \{(u, v) | 0 \leq u \leq 1, 0 \leq v \leq 1-u\}$,

注意到 $\left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| = 4uv$, 所以 $\iint_D (\sqrt{x} + \sqrt{y}) dx dy = 4 \int_0^1 du \int_0^{1-u} (u+v) uv dv = \frac{2}{15}$.

(3) $\iint_D (x-y^2)dxdy$, D 是由 $y=2$, $y^2-y-x=1$, $y^2+2y-x=2$ 所围成的平面区域.

解: 令 $\begin{cases} u=y^2-x \\ v=y, \end{cases}$ 则积分区域 D 在新坐标系下变换为

$$\Omega = \{(u, v) | 2-2v \leq u \leq v+1, \frac{1}{3} \leq v \leq 2\}.$$

注意到 $\left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| = 1$, 所以 $\iint_D (x-y^2)dxdy = -\int_{\frac{1}{3}}^2 dv \int_{2-2v}^{1+v} udu = -\frac{175}{54}$.

(4) $\iint_D (x+y)\sin(x-y)dxdy$, $D = \{(x, y) | 0 \leq x+y \leq \pi, 0 \leq x-y \leq \pi\}$.

解: 令 $x = \frac{1}{2}(u+v)$, $y = \frac{1}{2}(u-v)$, 则 $D' = \{(u, v) | 0 \leq u \leq \pi, 0 \leq v \leq \pi\}$, $\left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| = \frac{1}{2}$.

于是

$$\iint_D (x+y)\sin(x-y)dxdy = \iint_{D'} u \sin v \cdot \frac{1}{2} dudv = \frac{1}{2} \int_0^\pi udu \int_0^\pi \sin vdv = \frac{1}{2}\pi^2.$$

(5) $\iint_D e^{\frac{y}{x+y}}dxdy$, $D = \{(x, y) | x+y \leq 1, x \geq 0, y \geq 0\}$.

解: 令 $x=v-u$, $y=u$, 则 $D' = \{(u, v) | 0 \leq u \leq v, 0 \leq v \leq 1\}$, $\left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| = 1$.

于是 $\iint_D e^{\frac{y}{x+y}}dxdy = \iint_{D'} e^{\frac{u}{v}}dudv = \int_0^1 dv \int_0^v e^{\frac{u}{v}}du = \frac{1}{2}(e-1)$.

4. 解答证明题:

(1) 设 $f(x, y) \in C^2$ 且满足 $f(1, y) = 0$, $f(x, 1) = 0$, $\iint_D f(x, y)dxdy = a$, 其中

$$D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}. \text{ 计算二重积分 } \iint_D xyf''_{xy}(x, y)dxdy.$$

解: 因为 $f(1, y) = 0$, $f(x, 1) = 0$, 因此 $f'_y(1, y) = 0$, $f'_x(x, 1) = 0$. 这样

$$\begin{aligned} \iint_D xyf''_{xy}(x, y)dxdy &= \int_0^1 xdx \int_0^1 yf''_{xy}(x, y)dy = \int_0^1 x(yf'_x(x, y))\Big|_{y=0}^{y=1} - \int_0^1 f'_x(x, y)dy dx \\ &= -\int_0^1 dy \int_0^1 xf'_x(x, y)dx = \int_0^1 (xf(x, y))\Big|_0^1 + \int_0^1 f(x, y)dx dy \\ &= \int_0^1 \int_0^1 f(x, y)dxdy = a. \end{aligned}$$

(2) 记 $D = \{(x, y) \mid |x| \leq a, |y| \leq a\}$. 设 $f(x)$ 是连续偶函数, 证明:

$$\iint_D f(x-y) dx dy = 2 \int_0^{2a} (2a-u) f(u) du.$$

证明: 令 $\begin{cases} u = x-y \\ v = x+y. \end{cases}$ 则 $\begin{cases} x = \frac{1}{2}(u+v) \\ y = \frac{1}{2}(v-u) \end{cases}$ 且积分区域 $D = \{(x, y) \mid |x| \leq a, |y| \leq a\}$ 转化为新

坐标系下的区域 $D_1 = \{(u, v) \mid |u| + |v| \leq 2a\}$, 且 $\left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{2}$.

$$\text{故 } \iint_D f(x-y) dx dy = \frac{1}{2} \iint_{D_1} f(u) du dv = \frac{1}{2} 2 \int_0^{2a} du \int_{-2a+u}^{2a-u} f(u) dv = 2 \int_0^{2a} (2a-u) f(u) du.$$

(3) 设 $f(u)$ 连续, 且 $D = \{(x, y) \mid |x| \leq \frac{A}{2}, |y| \leq \frac{A}{2}\}$. 证明:

$$\iint_D f(x+y) dx dy = \int_{-A}^A f(u)(A-|u|) du.$$

证明: 令 $\begin{cases} u = x+y \\ v = y, \end{cases}$ 则积分区域 D 转化为新坐标系下的如下区域

$$\Omega = \{(u, v) \mid -\frac{A}{2} \leq v \leq u + \frac{A}{2}, -A \leq u \leq 0\} \cup \{(u, v) \mid u - \frac{A}{2} \leq v \leq \frac{A}{2}, 0 \leq u \leq A\},$$

注意到 $\left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| = 1$, 所以

$$\begin{aligned} \iint_D f(x+y) dx dy &= \int_{-A}^0 du \int_{-\frac{A}{2}}^{\frac{A}{2}+u} f(u) dv + \int_0^A du \int_{u-\frac{A}{2}}^{\frac{A}{2}} f(u) dv \\ &= \int_{-A}^0 f(u)(u+A) du + \int_0^A f(u)(A-u) du \\ &= \int_{-A}^A f(u)(A-|u|) du. \end{aligned}$$

证毕.

(4) 记 $D_\delta = \{(x, y) \mid \delta^2 \leq x^2 + y^2 \leq 1\}$. 设 $f(x, y) \in C^1$ 满足当 $x^2 + y^2 = 1$ 时, 有

$$f(x, y) = 0. \text{ 证明: } \lim_{\delta \rightarrow 0^+} \iint_{D_\delta} \frac{xf'_x(x, y) + yf'_y(x, y)}{x^2 + y^2} dx dy = -2\pi f(0, 0).$$

证明: 令 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$ 并记 $u(r, \theta) = f(r \cos \theta, r \sin \theta)$. 则

$$u'_r(r, \theta) = \cos \theta f'_x + \sin \theta f'_y = \frac{1}{r} (x f'_x(x, y) + y f'_y(x, y)).$$

$$\begin{aligned} \text{故 } \iint_{D_\delta} \frac{x f'_x(x, y) + y f'_y(x, y)}{x^2 + y^2} dx dy &= \iint_{D_\delta} \frac{r u'_r(r, \theta)}{r^2} r dr d\theta = \iint_{D_\delta} u'_r(r, \theta) dr d\theta \\ &= \int_0^{2\pi} d\theta \int_\delta^1 u'_r(r, \theta) dr = \int_0^{2\pi} (u(1, \theta) - u(\delta, \theta)) d\theta \\ &= \int_0^{2\pi} (f(\cos \theta, \sin \theta) - f(\delta \cos \theta, \delta \sin \theta)) d\theta \\ &= - \int_0^{2\pi} f(\delta \cos \theta, \delta \sin \theta) d\theta \\ &= -2\pi f(\delta \cos \varphi, \delta \sin \varphi), \end{aligned}$$

其中 $\varphi \in (0, 2\pi)$. 因为 $f(x, y) \in C^1$, 所以 $f(x, y)$ 在 $(0, 0)$ 连续, 从而

$$\lim_{\delta \rightarrow 0^+} \iint_{D_\delta} \frac{x f'_x(x, y) + y f'_y(x, y)}{x^2 + y^2} dx dy = -2\pi \lim_{\delta \rightarrow 0^+} f(\delta \cos \varphi, \delta \sin \varphi) = -2\pi f(0, 0).$$

(5) 设 $f(x, y) \in C^2$ 且关于两个变量 x 和 y 的周期都为 1, 即对任意的 (x, y) ,

$$f(x+1, y) = f(x, y), \quad f(x, y+1) = f(x, y). \quad \text{若 } f(x, y) \text{ 满足}$$

$$\int_{-1}^1 dx \int_{-1}^1 f(x, y) (f''_{xx}(x, y) + f''_{yy}(x, y)) dy \geq 0, \quad \text{证明: } f(x, y) \text{ 是常函数.}$$

证明: 因为

$$\begin{aligned} &\int_{-1}^1 dx \int_{-1}^1 f(x, y) (f''_{xx}(x, y) + f''_{yy}(x, y)) dy \\ &= \int_{-1}^1 dx \int_{-1}^1 f(x, y) f''_{xx}(x, y) dy + \int_{-1}^1 dx \int_{-1}^1 f(x, y) f''_{yy}(x, y) dy, \end{aligned}$$

而

$$\begin{aligned} &\int_{-1}^1 dx \int_{-1}^1 f(x, y) f''_{xx}(x, y) dy = \int_{-1}^1 dy \int_{-1}^1 f(x, y) f''_{xx}(x, y) dx \\ &= \int_{-1}^1 (f(x, y) f'_x(x, y) \Big|_{x=-1}^{x=1} - \int_{-1}^1 (f'_x(x, y))^2 dx) dy \\ &= - \int_{-1}^1 (\int_{-1}^1 (f'_x(x, y))^2 dx) dy \\ &= - \iint_{\substack{|x| \leq 1 \\ |y| \leq 1}} (f'_x(x, y))^2 dx dy, \end{aligned}$$

$$\begin{aligned}
& \int_{-1}^1 dx \int_{-1}^1 f(x, y) f_{yy}''(x, y) dy \\
&= \int_{-1}^1 (f(x, y) f_y'(x, y)) \Big|_{y=-1}^{y=1} - \int_{-1}^1 (f_y'(x, y))^2 dy dx \\
&= - \int_{-1}^1 \left(\int_{-1}^1 (f_y'(x, y))^2 dy \right) dx \\
&= - \iint_{\substack{|x| \leq 1 \\ |y| \leq 1}} (f_y'(x, y))^2 dx dy,
\end{aligned}$$

故当 $\int_{-1}^1 dx \int_{-1}^1 f(x, y)(f_{xx}''(x, y) + f_{yy}''(x, y)) dy \geq 0$ 时, 必有

$$\iint_{\substack{|x| \leq 1 \\ |y| \leq 1}} [(f_x'(x, y))^2 + (f_y'(x, y))^2] dx dy \leq 0,$$

由于 $(f_x'(x, y))^2 + (f_y'(x, y))^2$ 是非负连续函数, 因此对 $\forall (x, y)$ 满足 $|x| \leq 1, |y| \leq 1$, 有 $(f_x'(x, y))^2 + (f_y'(x, y))^2 = 0$. 从而对 $\forall (x, y)$ 满足 $|x| \leq 1, |y| \leq 1$, $f_x'(x, y) = 0$ 且 $f_y'(x, y) = 0$, 这样 $f(x, y)$ 在区域 $\{(x, y) \mid |x| \leq 1, |y| \leq 1\}$ 上是常数. 由函数 $f(x, y)$ 的周期性可知, $f(x, y)$ 在其定义域上是常数. 证毕

(6) 设 $f(x, y)$ 在开单位圆盘 $D = \{(x, y) \mid x^2 + y^2 < 1\}$ 上是 C^2 类函数, 在闭单位圆盘

$\bar{D} = \{(x, y) \mid x^2 + y^2 \leq 1\}$ 上连续. 若函数 $f(x, y)$ 在 $\partial D = \{(x, y) \mid x^2 + y^2 = 1\}$ 上取

值为常数零, 证明: $\iint_{x^2+y^2 \leq 1} f(x, y)[f_{xx}''(x, y) + f_{yy}''(x, y)] dx dy \leq 0$.

证明: 将重积分化为累次积分, 然后再做分部积分, 并利用假设条件.

$$\begin{aligned}
& \iint_{x^2+y^2 \leq 1} f(x, y) f_{xx}''(x, y) dx dy = \int_{-1}^1 dy \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f(x, y) f_{xx}''(x, y) dx = \int_{-1}^1 dy \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f df_x' \\
&= \int_{-1}^1 \left[f(x, y) f_x'(x, y) \Big|_{x=-\sqrt{1-y^2}}^{x=\sqrt{1-y^2}} - \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f_x'(x, y)^2 dx \right] dy = - \iint_{x^2+y^2 \leq 1} f_x'(x, y)^2 dx dy \leq 0.
\end{aligned}$$

同理可证 $\iint_{x^2+y^2 \leq 1} f(x, y) f_{yy}''(x, y) dx dy = - \iint_{x^2+y^2 \leq 1} f_y'(x, y)^2 dx dy \leq 0$. 因此

$$\iint_{x^2+y^2 \leq 1} f(x, y)[f_{xx}''(x, y) + f_{yy}''(x, y)] dx dy \leq 0. \text{ 证毕}$$

(7) 设 $f(x) \in C[0, 1]$ 且 $0 < m \leq f(x) \leq M$ ($\forall x \in [0, 1]$).

证明:
$$\iint_{\substack{0 \leq x \leq 1 \\ 0 \leq y \leq 1}} \frac{f(x)}{f(y)} dx dy \leq \frac{(M+m)^2}{4Mm}.$$

证明:
$$\iint_{\substack{0 \leq x \leq 1 \\ 0 \leq y \leq 1}} \frac{f(x)}{f(y)} dx dy = \int_0^1 \frac{1}{f(y)} dy \int_0^1 f(x) dx = \int_0^1 \frac{1}{f(x)} dx \int_0^1 f(x) dx.$$

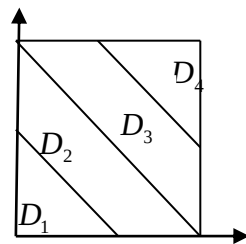
因为对 $\forall x \in [0, 1]$, $(M - f(x))(f(x) - m) \geq 0$, 因此 $(M + m)f(x) \geq f^2(x) + Mm$, 从而

$$M + m \geq f(x) + \frac{Mm}{f(x)}, \text{ 两边积分得, } M + m \geq \int_0^1 f(x) dx + Mm \int_0^1 \frac{1}{f(x)} dx,$$

记 $a = \int_0^1 f(x) dx$, $b = \int_0^1 \frac{1}{f(x)} dx$, 则 $M + m \geq a + bMm \geq 2\sqrt{abMm}$, 故 $ab \leq \frac{(M+m)^2}{4Mm}$.

所以
$$\iint_{\substack{0 \leq x \leq 1 \\ 0 \leq y \leq 1}} \frac{f(x)}{f(y)} dx dy = ab \leq \frac{(M+m)^2}{4Mm}.$$

(8) 求 $I = \iint_D [x+y] d\sigma$, 其中 $D = [0, 2] \times [0, 2]$, $[x+y]$ 为取整函数。



解: 为方便, 对 D 作分解 $D = D_1 \cup D_2 \cup D_3 \cup D_4$, 如图。于是

$$\begin{aligned} I &= \iint_D [x+y] d\sigma = \iint_{D_1} [x+y] d\sigma + \iint_{D_2} [x+y] d\sigma + \iint_{D_3} [x+y] d\sigma + \iint_{D_4} [x+y] d\sigma \\ &= \iint_{D_1} 0 d\sigma + \iint_{D_2} 1 \cdot d\sigma + \iint_{D_3} 2 \cdot d\sigma + \iint_{D_4} 3 \cdot d\sigma \\ &= S(D_2) + 2S(D_3) + 3S(D_4) = 6. \end{aligned}$$

其中 $S(D_2) = S(D_3) = \frac{3}{2}$, $S(D_1) = S(D_4) = \frac{1}{2}$, 解答完毕。

(9) 计算 $I = \iint_D \frac{1}{\sqrt{x^2 + y^2}} \left(y \frac{\partial f}{\partial x} - x \frac{\partial f}{\partial y} \right) dx dy$, 其中 $D = \{(x, y) | x^2 + y^2 \leq R^2\}$ 且

$$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \in C(D).$$

解: 考虑极坐标变换 $\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta, \end{cases} dx dy = \rho d\rho d\theta$. 则

$$\frac{1}{\sqrt{x^2+y^2}}\left(y\frac{\partial f}{\partial x}-x\frac{\partial f}{\partial y}\right)=\frac{1}{\rho}\frac{\partial f}{\partial(x,y)}\begin{pmatrix}y\\-x\end{pmatrix}=\frac{1}{\rho}\frac{\partial f}{\partial(\rho,\theta)}\cdot\frac{\partial(\rho,\theta)}{\partial(x,y)}\begin{pmatrix}y\\-x\end{pmatrix}=-\frac{1}{\rho}\frac{\partial f}{\partial\theta},$$

$$\begin{aligned}\text{其中}\frac{\partial(\rho,\theta)}{\partial(x,y)}\begin{pmatrix}y\\-x\end{pmatrix}&=\left(\frac{\partial(x,y)}{\partial(\rho,\theta)}\right)^{-1}\begin{pmatrix}y\\-x\end{pmatrix}=\begin{pmatrix}\cos\theta & -\rho\sin\theta\\ \sin\theta & \rho\cos\theta\end{pmatrix}^{-1}\begin{pmatrix}y\\-x\end{pmatrix} \\ &=\frac{1}{\rho}\begin{pmatrix}\rho\cos\theta & \rho\sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}y\\-x\end{pmatrix}=\frac{1}{\rho}\begin{pmatrix}0\\0\end{pmatrix}=\begin{pmatrix}0\\-1\end{pmatrix}.\end{aligned}$$

$$\text{故 } I = \iint_D \frac{1}{\sqrt{x^2+y^2}} \left(y \frac{\partial f}{\partial x} - x \frac{\partial f}{\partial y} \right) dx dy = - \iint_{\substack{0 \leq \rho \leq R \\ 0 \leq \theta \leq 2\pi}} \frac{1}{\rho} \frac{\partial f}{\partial \theta} \rho d\rho d\theta$$

$$= - \int_0^R d\rho \int_0^{2\pi} \frac{\partial f}{\partial \theta} d\theta = - \int_0^R (f(\rho, 2\pi) - f(\rho, 0)) d\rho = 0.$$

$$(10) \text{ 计算 } I = \iint_D |x^2 + y^2 - 4| d\sigma, \quad D = \{(x, y) | x^2 + y^2 \leq 16\}.$$

解: 记 $D = D_1 \cup D_2$, $D_1 = \{(x, y) | x^2 + y^2 \leq 4\}$, $D_2 = \{(x, y) | 4 \leq x^2 + y^2 \leq 16\}$, 则

$$\begin{aligned}I &= \iint_{D_1} (4 - x^2 - y^2) dx dy + \iint_{D_2} (x^2 + y^2 - 4) dx dy = \int_0^{2\pi} d\theta \int_0^2 (4 - r^2) r dr + \int_0^{2\pi} d\theta \int_2^4 (r^2 - 4) r dr \\ &= 2\pi \left(2r^2 - \frac{r^4}{4} \right) \Big|_0^2 + 2\pi \left(\frac{r^4}{4} - 2r^2 \right) \Big|_2^4 = 80\pi.\end{aligned}$$

$$\begin{aligned}\text{或者 } I &= - \iint_{D_1} (x^2 + y^2 - 4) dx dy + \iint_{D_2} (x^2 + y^2 - 4) dx dy \\ &= \iint_{D_1 \cup D_2} (x^2 + y^2 - 4) dx dy - 2 \iint_{D_1} (x^2 + y^2 - 4) dx dy \\ &= \int_0^{2\pi} d\theta \int_0^4 (r^2 - 4) r dr - 2 \int_0^{2\pi} d\theta \int_0^2 (r^2 - 4) r dr = 80\pi.\end{aligned}$$

解答完毕

5. 利用二重积分理论, 证明下列结论: 设 $f(x)$, $g(x)$ 在 $[a, b]$ 上连续, 则

$$(1) \left(\int_a^b f(x) dx \right)^2 \leq (b-a) \int_a^b f^2(x) dx;$$

$$(2) \left(\int_a^b f(x) g(x) dx \right)^2 \leq \int_a^b f^2(x) dx \int_a^b g^2(x) dx.$$

$$(3) \int_a^b dx \int_x^b f(x) f(y) dy = \frac{1}{2} \left(\int_a^b f(x) dx \right)^2.$$

证明:

$$\begin{aligned}
 (1) \quad & \left(\int_a^b f(x) dx \right)^2 = \int_a^b f(x) dx \int_a^b f(y) dy = \iint_{[a,b] \times [a,b]} f(x) f(y) dx dy \\
 & \leq \frac{1}{2} \iint_{[a,b] \times [a,b]} [f^2(x) + f^2(y)] dx dy = \frac{1}{2} \iint_{[a,b] \times [a,b]} f^2(x) dx dy + \frac{1}{2} \iint_{[a,b] \times [a,b]} f^2(y) dx dy \\
 & = \frac{1}{2} (b-a) \int_a^b f^2(x) dx + \frac{1}{2} (b-a) \int_a^b f^2(y) dy = (b-a) \int_a^b f^2(x) dx.
 \end{aligned}$$

(2) 由不等式 $[f(x)g(y) - f(y)g(x)]^2 \geq 0$ 得

$$\begin{aligned}
 0 & \leq \iint_{[a,b] \times [a,b]} [f(x)g(y) - f(y)g(x)]^2 dx dy \\
 & = \iint_{[a,b] \times [a,b]} [f^2(x)g^2(y) + f^2(y)g^2(x) - 2f(x)g(x)f(y)g(y)] dx dy \\
 & = 2 \int_a^b f^2(x) dx \int_a^b g^2(x) dx - 2 \left(\int_a^b f(x)g(x) dx \right)^2. \text{ 由此立刻得到不等式(2).}
 \end{aligned}$$

(3) 令 $D = \{(x, y) \mid a \leq x \leq b, x \leq y \leq b\}$, $E = \{(x, y) \mid a \leq x \leq b, a \leq y \leq x\}$. 则

D 与 E 关于 $y = x$ 对称, 因此

$$\int_a^b dx \int_x^b f(x) f(y) dy = \frac{1}{2} \int_a^b f(x) dx \int_a^b f(y) dy = \frac{1}{2} \left(\int_a^b f(x) dx \right)^2.$$

(4) 若 $f(x)$ 是 $[a, b]$ 上的非负连续函数, 则

$$\left(\int_a^b f(x) \cos kx dx \right)^2 + \left(\int_a^b f(x) \sin kx dx \right)^2 \leq \left(\int_a^b f(x) dx \right)^2.$$

证明: 将不等式左边的每一项改写为二重积分,

$$\left(\int_a^b f(x) \cos kx dx \right)^2 = \int_a^b \int_a^b f(x) f(y) \cos kx \cos ky dx dy,$$

$$\left(\int_a^b f(x) \sin kx dx \right)^2 = \int_a^b \int_a^b f(x) f(y) \sin kx \sin ky dx dy,$$

由于 $\cos k(x-y) = \cos kx \cos ky + \sin kx \sin ky$, 并注意到, $f(y) \cos k(x-y) \leq f(y)$,

所以

$$\begin{aligned} & \left(\int_a^b f(x) \cos kx dx \right)^2 + \left(\int_a^b f(x) \sin kx dx \right)^2 = \int_a^b \int_a^b f(x) f(y) \cos k(x-y) dx dy \\ & = \int_a^b f(x) dx \int_a^b f(y) \cos k(x-y) dy \leq \left(\int_a^b f(x) dx \right)^2. \end{aligned}$$

(5) 若 $f(x)$, $p(x)$, $g(x)$ 在 $[a, b]$ 上连续, $p(x)$ 是正值函数, $f(x)$, $g(x)$ 都是单调增加函数或都是单调减小函数, 证明:

$$\int_a^b p(x) f(x) dx \cdot \int_a^b p(x) g(x) dx \leq \int_a^b p(x) dx \cdot \int_a^b p(x) f(x) g(x) dx.$$

(此不等式称为切比雪夫不等式)

证明: 由于定积分与积分变量无关, 将所证不等式中的积分乘积化为二重积分, 考虑差

$$\begin{aligned} \Delta &= \int_a^b p(x) dx \cdot \int_a^b p(x) f(x) g(x) dx - \int_a^b p(x) f(x) dx \cdot \int_a^b p(x) g(x) dx \\ &= \int_a^b p(x) f(x) g(x) dx \cdot \int_a^b p(y) dy - \int_a^b p(x) f(x) dx \cdot \int_a^b p(y) g(y) dy \\ &= \iint_D (p(x) f(x) g(x) p(y) - p(x) f(x) p(y) g(y)) dx dy \\ &= \iint_D p(x) f(x) p(y) (g(x) - g(y)) dx dy, \end{aligned}$$

由于积分区域 $D = [a, b] \times [a, b]$ 关于直线 $y = x$ 对称, 因此由轮换对称性知,

$$\begin{aligned} \Delta &= \iint_D p(x) f(x) p(y) (g(x) - g(y)) dx dy \\ &= \iint_D p(y) f(y) p(x) (g(y) - g(x)) dx dy \\ &= \frac{1}{2} \iint_D p(x) p(y) (f(x) - f(y)) (g(x) - g(y)) dx dy, \end{aligned}$$

因为 $p(x)$ 是 $[a, b]$ 的正值函数, $f(x)$, $g(x)$ 在 $[a, b]$ 上的单调性相同, 于是, 在 D 上, 被积函数

$$p(x) p(y) (f(x) - f(y)) (g(x) - g(y)) \geq 0,$$

从而 $\Delta \geq 0$, 即 $\int_a^b p(x) f(x) dx \cdot \int_a^b p(x) g(x) dx \leq \int_a^b p(x) dx \cdot \int_a^b p(x) f(x) g(x) dx$. 证毕

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以下内容为学有余力的同学选做。

6. 设函数 $f(x, y)$ 及其偏导数 $f'_y(x, y)$ 在平面区域 D 上连续, 其中

$D = \{(x, y) | a \leq x \leq b, \varphi(x) \leq y \leq \psi(x)\}$, 这里 $\varphi(x)$ 和 $\psi(x)$ 为 $[a, b]$ 上的连续函数, 且

$\varphi(x) \leq \psi(x)$. 进一步假设 $f(x, \varphi(x)) = 0, \forall x \in [a, b]$. 证明存在常数 $C > 0$, 使得

$$\iint_D f^2(x, y) dx dy \leq C \iint_D (f'_y(x, y))^2 dx dy. \quad (\text{这个不等式称作 Poincare 不等式})$$

证明: 根据假设和 Newton—Leibniz 公式得 $f(x, y) = \int_{\varphi(x)}^y f'_t(x, t) dt$.

两边平方并应用 Cauchy-Schwarz 不等式得

$$f^2(x, y) = \left(\int_{\varphi(x)}^y f'_t(x, t) dt \right)^2 \leq (y - \varphi(x)) \int_{\varphi(x)}^y (f'_t(x, t))^2 dt \leq [\psi(x) - \varphi(x)] \int_{\varphi(x)}^{\psi(x)} (f'_y(x, y))^2 dy$$

两边关于 y 在区间 $[\varphi(x), \psi(x)]$ 上积分得

$$\int_{\varphi(x)}^{\psi(x)} f^2(x, y) dy \leq [\psi(x) - \varphi(x)]^2 \int_{\varphi(x)}^{\psi(x)} (f'_y(x, y))^2 dy.$$

记 $C = \max\{[\psi(x) - \varphi(x)]^2, a \leq x \leq b\}$. 则 $\int_{\varphi(x)}^{\psi(x)} f^2(x, y) dy \leq C \int_{\varphi(x)}^{\psi(x)} (f'_y(x, y))^2 dy$.

对上述不等式关于 x 在区间 $[a, b]$ 上积分得 $\int_a^b dx \int_{\varphi(x)}^{\psi(x)} f^2(x, y) dy \leq C \int_a^b dx \int_{\varphi(x)}^{\psi(x)} (f'_y(x, y))^2 dy$.

再将上式两边的累次积分换成重积分, 即得所要证明的 Poincare 不等式. 证毕

7. 计算 $I = \iint_D \left| \frac{x+y}{\sqrt{2}} - x^2 - y^2 \right| d\sigma$, 其中 D 为闭单位圆域 $x^2 + y^2 \leq 1$.

解: 为了计算积分, 应先去掉被积函数的绝对值符号. 记 $f(x, y) = \frac{x+y}{\sqrt{2}} - x^2 - y^2$.

将函数 $f(x, y)$ 表示为如下形式

$$f(x, y) = \frac{1}{4} - \left(x - \frac{1}{2\sqrt{2}} \right)^2 - \left(y - \frac{1}{2\sqrt{2}} \right)^2. \quad \text{记 } D_1 = \{(x, y) \mid f(x, y) \geq 0\},$$

即 D_1 是以点 $\left(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}} \right)$ 为圆心, 半径为 $1/2$ 的闭圆盘. 注意 D_1 完全包含在闭单位圆盘 D

之中. 于是 $I = \iint_{D_1} f(x, y) dx dy - \iint_{D \setminus D_1} f(x, y) dx dy = 2 \iint_{D_1} f(x, y) dx dy - \iint_D f(x, y) dx dy$.

对积分 $\iint_{D_1} f(x, y) dx dy$, 做变换 $x = \frac{1}{2\sqrt{2}} + r \cos \vartheta, y = \frac{1}{2\sqrt{2}} + r \sin \vartheta$, 则

$$\iint_{D_1} f(x, y) dx dy = \iint_{0 \leq r \leq 1/2, 0 \leq \vartheta \leq 2\pi} \left(\frac{1}{4} - r^2 \right) r dr d\vartheta = 2\pi \int_0^{1/2} \left(\frac{r}{4} - r^3 \right) dr = \frac{\pi}{32}.$$

对积分 $\iint_D f(x, y) dx dy$, 做极坐标变换 $x = r \cos \vartheta$, $y = r \sin \vartheta$ 得

$$\iint_D f(x, y) dx dy = \iint_{0 \leq r \leq 1, 0 \leq \vartheta \leq 2\pi} \left(-r^2 + \frac{r(\cos \vartheta + \sin \vartheta)}{\sqrt{2}} \right) r dr d\vartheta = -\frac{\pi}{2}$$

故所求积分为 $I = \frac{2\pi}{32} + \frac{\pi}{2} = \frac{9\pi}{16}$. 解答完毕

8. 计算 $I = \iint_D |\cos(x+y)| d\sigma$, 其中 D 为闭正方形: $0 \leq x \leq \pi$, $0 \leq y \leq \pi$.

解: 做变换 $\varphi: u = x+y$, $v = y$. 其逆变换为 $x = u-v$, $y = v$. 不难看出线性变换 φ 将 xy 平面上的闭正方形 D , 变换成 uv 平面上的闭平行四边形 Ω , 平行四边形 Ω 由四条直线 $v=0$, $v=\pi$, $u-v=0$, $u-v=\pi$ 所围成的。由于线性变换 φ 的 Jacobian 行列式是 1, 故所求积分可表为 $I = \iint_{\Omega} |\cos u| d\sigma$. 为计算方便, 我们将平行四边形 Ω 可表为两个三角形

之并 $\Omega = \Omega_1 \cup \Omega_2$, 其中 $\Omega_1 = \{(u, v), 0 \leq u \leq \pi, 0 \leq v \leq u\}$,

$\Omega_2 = \{(u, v), \pi \leq u \leq 2\pi, u \leq v \leq \pi\}$.

于是 $I = \iint_{\Omega_1} |\cos u| du dv + \iint_{\Omega_2} |\cos u| du dv$.

$$\text{其中 } \iint_{\Omega_1} |\cos u| du dv = \int_0^{\pi/2} \cos u \int_0^u dv - \int_{\pi/2}^{\pi} \cos u du \int_0^u dv = \int_0^{\pi/2} u \cos u du - \int_{\pi/2}^{\pi} u \cos u du$$

$$= u \sin u \Big|_0^{\pi/2} - u \sin u \Big|_{\pi/2}^{\pi} = \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$

$$\iint_{\Omega_2} |\cos u| du dv = \int_{\pi}^{3\pi/2} -\cos u \int_u^{\pi} dv + \int_{3\pi/2}^{2\pi} \cos u du \int_u^{\pi} dv = \int_{\pi}^{3\pi/2} (u-\pi) \cos u du + \int_{3\pi/2}^{2\pi} (\pi-u) \cos u du$$

$$= \int_0^{\pi/2} s \cos s ds - \int_{\pi/2}^{\pi} t \cos t dt = \pi. \text{ 故所求积分为 } I = 2\pi. \text{ 解答完毕。}$$

解法二、由于积分区域关于 $x+y=\pi$ 对称, 被积函数在对称点上的函数值相等, 因此

$$I = \iint_D |\cos(x+y)| d\sigma = 2 \iint_{D_1} |\cos(x+y)| d\sigma, \text{ 其中 } D_1 = \{(x, y) | 0 \leq x \leq \pi, 0 \leq y \leq \pi-x\}.$$

注意到, $\frac{\pi}{2} \leq x+y \leq \pi$ 时, $\cos(x+y) \leq 0$, 且被积函数 $|\cos(x+y)|$ 在下列两个区域上的

积分相等: 由 x 轴, y 轴及直线 $x+y=\frac{\pi}{2}$ 围成的区域及由 x 轴, 直线 $x=\pi$ 及直线 $x+y=\frac{\pi}{2}$ 围成的区域。因此 $\iint_{D_1} |\cos(x+y)| d\sigma = \iint_{D_4} |\cos(x+y)| d\sigma$, 其中

$$D_4 = \{(x, y) \mid 0 \leq x \leq \pi, \frac{\pi}{2} - x \leq y \leq \pi - x\}, \text{ 这样}$$

$$I = \iint_D |\cos(x+y)| d\sigma = 2 \iint_{D_4} |\cos(x+y)| d\sigma = 2 \int_0^\pi dx \int_{\frac{\pi-x}{2}}^{\pi-x} -\cos(x+y) dy = 2\pi.$$

以下部分内容大纲不做要求:

二重积分的积分区域和被积函数都是有界的, 将有界区域推广到无界区域, 就有无穷二重积分, 将有界函数推广到无界函数, 就有瑕二重积分, 无穷二重积分和瑕二重积分统称为广义二重积分, 下面只给出无穷二重积分收敛与发散的概念, 瑕二重积分收敛与发散的概念可类似瑕积分写出。

定义: 若函数 $f(x, y)$ 定义在无界区域 D 上, 符号 $\iint_D f(x, y) dx dy$ 称为无穷二重积分. 如果

任意包含原点的有界区域 G , 函数 $f(x, y)$ 在 $G \cap D = E$ 上可积, 设

$$d_G = \min\{\sqrt{x^2 + y^2} \mid (x, y) \in \partial G \text{ (区域 } G \text{ 的边界)}\}.$$

若极限 $\lim_{d_G \rightarrow +\infty} \iint_E f(x, y) dx dy$ 存在, 称无穷二重积分 $\iint_D f(x, y) dx dy$ 收敛, 该极限称为函数

$f(x, y)$ 在无界区域 D 上的积分, 且 $\iint_D f(x, y) dx dy = \lim_{d_G \rightarrow +\infty} \iint_E f(x, y) dx dy$. 若极限不存在, 称无穷二重积分 $\iint_D f(x, y) dx dy$ 发散.

9. 计算二重广义积分 $\iint_{R^2} e^{-(x^2+y^2)} \sin(x^2 + y^2) dx dy$.

解: 作极坐标变换: $x = r \cos t$, $y = r \sin t$, 则所求积分为

$$\iint_{0 \leq r < +\infty, 0 \leq t \leq 2\pi} e^{-r^2} \sin r^2 r dr dt = \frac{1}{2} \int_0^{2\pi} dt \int_0^{+\infty} e^{-s} \sin s ds$$

注意 $\int_0^{+\infty} e^{-s} \sin s ds = \frac{-e^{-s}(\cos s + \sin s)}{2} \Big|_{s=0}^{s=+\infty} = \frac{1}{2}$. 于是我们得到

$$\iint_{R^2} e^{-(x^2+y^2)} \sin(x^2 + y^2) dx dy = \frac{\pi}{2}. \text{ 解答完毕}$$

10. 计算二重广义积分 $\iint_{R^2} e^{2xy-2x^2-y^2} dx dy$.

解：注意 $2xy - 2x^2 - y^2 = -(x - y)^2 - x^2$. 令 $u = x$, $v = x - y$, 则

其逆变换为 $x = u$, $y = u - v$. 于是原积分等于

$$\iint_{\mathbb{R}^2} e^{2xy - 2x^2 - y^2} dx dy = \iint_{\mathbb{R}^2} e^{-u^2 - v^2} \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du dv = \int_{-\infty}^{+\infty} e^{-u^2} du \int_{-\infty}^{+\infty} e^{-v^2} dv = \pi. \text{ 解答完毕}$$